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Finding structure in noise

What repeated measurements can tell us

Your name

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How precisely can we measure the signal?

One measurement

$$x_i = \mu + \varepsilon_i$$

A fixed signal μ , disturbed by zero-mean noise ε_i .

The sample mean

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$$

How much uncertainty remains after averaging N measurements?

Averaging reduces independent noise

$$\text{SE}(\hat{\mu}) = \frac{\sigma}{\sqrt{N}}$$

Independent samples, each with variance σ^2 .

Their variances add: $\text{Var}(\hat{\mu}) = \frac{\sigma^2}{N}$.

Four times the samples halves the error

Standard error / σ

$N = 1$		1.00
$N = 4$		0.50
$N = 16$		0.25

Model prediction for independent samples.

Each twofold improvement in precision costs four times as many samples.

More data helps, with diminishing returns.

Shared errors can survive averaging

Independent errors

$$\text{SE}(\hat{\mu}) = \frac{\sigma}{\sqrt{N}}$$

Different errors cancel in the average.

Identical errors

$$\text{SE}(\hat{\mu}) = \sigma$$

Every sample carries the same error.

Identical errors are the limiting case of perfect positive correlation.

**Check the noise before
collecting more samples.**